



CS-E4960 - Software Testing and Quality Assurance D, Lecture, 3.9.2024-26.11.2024

This course space end date is set to 26.11.2024 **Search Courses:** **CS-E4960**

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Static Code Analysis 1: Pylint

To do: Make a submission

To do: Receive a grade

Opened: Tuesday, 22 October 2024, 10:00 AM
Due: Tuesday, 29 October 2024, 10:00 AM

Instructions

Imagine yourself as a software developer working for a company that has established its own coding standards. Among others, all of the company's Python code needs to comply with the following rules:

- functions and methods should not have more than 4 arguments
- there should not be more than three boolean statements in an if condition
- maximum number of nested blocks for a function or method body is 3.
- all lines in a module should have at most 80 characters
- there are no unused variables

So far, the company has enforced its coding standards only informally during manual code reviews, which has become too exhausting and resource-intensive due to the increasing complexity of the codebase

Therefore, your task is to enable automated checking of the company coding standards by creating a custom configuration for the Pylint tool. The configuration must check compliance with all the rules listed above. Using your configuration, Pylint should report only violations of these rules. Other checks should be disabled in the configuration.

Submission

Submit a zip archive containing:

- config.txt file with the pylint configuration
- sample-fail.py file with sample code that violates the specified rules
- sample-pass.py file with sample code that is compliant with the specified rules
- output-pass.txt with the pylint analysis output for the sample-pass.py
- output-fail.txt with the pylint analysis output for the sample-fail.py

Notes:

You can generate a default config file by `pylint --generate-rcfile > config.txt`

The generated config contains lots of options, which make it difficult to read. You can use it as a base for your config, but make sure to remove all options that are not required for the task.

To run Pylint with your configuration, you can use `pylint --rcfile=config.txt file_to_check.py`

Use the pylint documentation to look for options related to the specified rules and general information about tool's usage: <https://pylint.readthedocs.io/en/latest/index.html>

You can enable/disable messages in the `[MESSAGE CONTROL]` section of the config file. Also note that to enable reporting of checks that are specified by the assignment, you need to use the correct message identifier, which is usually different from the check name. The list of message identifiers can be found at: https://pylint.pycqa.org/en/latest/user_guide/messages/messages_overview.html

Grading:

To receive full points, you must submit all required files, implement required Pylint checks and provide Pylint configuration according to the assignment requirements. Partial completion will result in partial credit.

Use of AI and Collaboration with Other Students: Using AI tools to produce an answer for this assignment is not permitted.All parts of the submission must be produced by yourself (apart from files provided in the assignment). Collaboration with other students is not permitted. This is an individual assignment that you must author independently.

Submission status

Attempt number	This is attempt 1.
Submission status	No submissions have been made yet
Grading status	Not marked
Time remaining	Assignment is overdue by: 134 days 3 hours
Last modified	-

Previous activity

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Next activity

Static Code Analysis 2: SonarQube ►



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